

Version History

Complete development history of OQUL EdTech OS from foundation through Phase 55 production release.



55
Build Phases



2024–26
Development Period



17
DB Migrations



v1.0
Current Release

Release Timeline

0

Phase 0 — Foundation

Project architecture established. Database schema designed. Environment configuration standardized. Drizzle ORM selected over Prisma for performance.

Foundation · ADR-001 through ADR-004

1

Phases 1–3 — Core Infrastructure

Database architecture finalized (PostgreSQL + Drizzle). Authentication system built (JWT + HTTP-only cookies). Application runtime established.

Infrastructure Layer

4

Phases 4–8 — Product APIs & Runtime

Product API contracts defined. AI provider abstraction layer built. Real-time AI streaming via Groq API. Production hardening for AI reliability. Rate limiting implemented.

Product Runtime · ADR-005

14

Phase 14 — Living Learning System

First adaptive learning migration. Student knowledge state tracking introduced. Redis caching layer for learning data.

Migration 014

27

Phase 27 — Living Learner Intelligence

Learner memory system. Personalized learning paths. Confidence scoring engine. Student emotional state modeling.

Migration 015

28

Phase 28 — Human Learning Experience

Emotional memory integration. Daily challenge system. Human-centered UX patterns. Leila persona deepened with empathy layers.

Migration 016 · Emotional Intelligence

29

Phase 29 — Adaptive Learning Intelligence

Knowledge tracing algorithms. Bayesian knowledge estimation. Cognitive pacing system. Learning velocity measurement.

38

Phase 38 — Stabilization & Security

Security hardening. Rate limiting refinement. Error boundary improvements. Child safety compliance (COPPA-aligned checks).

Security & Compliance

40

Phase 40 — Learning Experience Core

Core learning UX rebuilt. Gamification system (XP, streaks, badges). Adventure map journey system. Engagement mechanics.

UX & Gamification

42

Phase 42 — Middle School Intelligence

Middle school curriculum engine. Grade 7–9 adaptive paths. Subject-specific AI tutor behaviors for Arabic and Mathematics.

Middle School Module

50

Phases 50–53 — Full Curriculum System

Complete K–12 curriculum mapping. Primary school (Phases 52), middle school (Phase 51), secondary school (Phase 53, TC/1BAC/2BAC). Deep content enhancement system.

Complete Curriculum Registry

54

Phase 54 — Official Alignment & Exam Intelligence

Alignment to official Moroccan Ministry of Education standards. Competency matrix for all subjects. Exam-ready assessment system.

Official Curriculum Alignment

55

Phase 55 — Adaptive AI Learning Platform**Current**

Full adaptive AI platform. Phase 55 adaptive platform feature module. Production hardening complete. Load testing verified. Parent portal. Complete observability stack.

Production Release v1.0 · 2026

**What's Coming Next**

See [06_META/Product_Roadmap.pdf](#) for the Phase 56+ feature roadmap including content factory automation, multi-language support, and B2B school licensing.